Fiscal Year:	FY 2006	**Task Last Updated:**	FY 06/25/2007
PI Name:	Ramsey, John Michael Ph.D.		
Project Title:	Automated Microfluidic Devices for Monitoring Biological Systems in Space		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Biotechnology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Chapel Hill	**State:**	NC
Zip Code:	27599	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	08/01/2001	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	4	**No. of Master' Degrees:**	1
No. of Master's Candidates:	5	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/30/2006 per P. Denkins/JSC (5/2007)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gonda, Steven (NASA Johnson Space Center)		
Grant/Contract No.:	NNJ05HB38G		
Performance Goal No.:			
Performance Goal Text:			
	Exposure to the unique environmental conditions found in space (i.e., low gravity, potentially toxic chemicals in enclosed spaces, and ionizing radiation) raises the prospects for significant biological consequences arising from radiation and chemical-induced damage at the genetic level compounded by the potentially synergistic effects of microgravity. To better understand these effects, a collaborative effort between the Ramsey group at The University of North Carolina and NASA's Johnson Space Center has been established to develop robust biosensers for space applications. These efforts seek to integrate the lab-on-a-chip technology developed at ORNL with space bioreactors developed at NASA. The lab-on-a-chip technology will be used both to monitor the performance of the bioreactor and to detect and quantitate DNA changes in cells cultured in the bioreactor By coupling the reporter organisms to an analytical measuring instrument, specific biomolecules can be detected and monitored as an indicator of either bioreactor performance on the International Space Station or the effect of environmental insults at planetary surfaces. This		

Task Description:	biosensor technology will be developed and tested for low-gravity operation and will be adaptable to a wide range of cell types. The entire system will be automated so that no human intervention will be required while on orbit. In addition, because of the compact nature of the lab-on-a-chip devices they can be integrated with the present bioreactor systems with minimal increases in space, power, and mass allotment. Our objective is to develop an enabling biosensor technology that can be effectively utilized for 1) the routine monitoring and detection of a wide variety of molecular compounds and 2) the identification of characteristic chemical signatures produced by living organisms as a result of specific environmental stresses. This technology will integrate the lab-on-a-chip devices (microchips) with space bioreactors. The first steps in realizing this goal will be 1) to develop a prototype microfluidic device capable of operating in reduced gravity and 2) to develop assays to detect changes/mutations in genes in response to environmental stressors. The micro-biosensors developed through this collaboration offer the capability of monitoring and determining the status of many systems, the levels of specific components within systems, and changes in biological organisms. This technology is critical to assessing the environment of spacecraft and planetary habitats. It has the potential ability to probe multiple events in near real time at both the genetic and the molecular level, enabling an assessment of the alterations in the regulatory genetic and molecular pathways associated with normal and altered growth of microorganisms and mammalian cells and tissues due to environmental stressors (radiation, toxic environments, microgravity). By combining lab-on-a-chip devices with their powerful capability for the automated monitoring of a variety of target molecules (analytes) with minimal power, space and mass requirements with a referenced set of bioreporters cells, the proposed biosensor system will be an enabling technology in support of the objectives and goals of long-duration missions of the HEDS program. The proposed research will bring a novel and powerful tool to space exploration and will provide answers to critical questions of how terrestrial life will fare in the extreme environments in space and on planetary surfaces.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The small, portable and robust biosentinel systems developed through the integration of NASA's bioreactor technology and Lab-on-a-Chip technology have applications far beyond those of interest to NASA for assuring the safety of its astronauts in hostile extraterrestrial environments. These systems can also be used for the early detection of a biological, nuclear, or chemical attacks on a building, transportation hub (i.e. subway or train station, airport, or seaport), city or military installation. They also can be applied to medical research. Such devices could be used to study cancer or Alzheimer etiology and also the effects of radiation and other chemical or environmental stressors on human beings at the cellular level.
Task Progress:	This grant focused on developing a microfluidic-based system capable of sampling cell culture effluent under microgravity conditions. Over the period of the grant we successfully designed, built, and tested a portable microfluidic system capable of performing rapid electrophoretic separations under a variety of gravitational conditions. This instrument was successfully flown multiple times on NASA's microgravity research aircraft as reported in the 2005 Analytical Chemistry publication listed below. A second generation of the lab-on-a-chip instrument was designed with an integrated interface to a NASA flight-certified cell culture bag. This interface was successfully tested on earth and in the microgravity research aircraft using cell media that was spiked with both fluorescently labeled amino acids and green fluorescent protein (GFP). A summary of this part of the research is presently being prepared for publication. Our aim in this part of the research was to demonstrate the ability of microfabricated devices to repeatedly and reproducibly sample microliter volumes from flight certified cell culture bags and perform capillary electrophoretic separations in 0, lunar, Martian, and hypergravity (1.8g).
Bibliography Type:	Description: (Last Updated: 06/25/2007)
Articles in Peer-reviewed Journals	Culbertson CT, Tugnawat Y, Meyer AR, Roman GT, Ramsey JM, Gonda SR. "Microchip separations in reduced-gravity and hypergravity environments." Anal Chem. 2005 Dec 15;77(24):7933-40. PMID: 16351140 http://dx.doi.org/10.1021/ac051198e , Dec-2005
Articles in Peer-reviewed Journals	Culbertson CT, Meyer AR, Roman GT, Ramsey JM. "Interfacing Cell Culture Arrays with Microfluidic Devices in Microgravity." in preparation. June 2007. , Jun-2007
Significant Media Coverage	Mukhopadhyay R. "Research highlights: Prepping microchips for infinity and beyond." Anal Chem A-Pages 2006 Jan 1;78(1):10. http://pubs.acs.org/subscribe/journals/ancham-a/78/i01/pdf/0106profile1.pdf , Jan-2006